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PALAEOMAGNETIC AND GEOCHEMICAL STUDIES ON DOLERITE DYKES FROM TAMIL NADU, INDIA

A.S. VENKATESH, G.V.S. POORNACHANDRA RAO, N.T.V. PRASADA RAO and M.S. BHALLA

Palaeomagnetic Group, National Geophysical Research Institute, Uppal Road, Hyderabad — 500 007 (India)

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ABSTRACT

Venkatesh, A.S., Poornachandra Rao, G.V.S., Prasada Rao, N.T.V. and Bhalla, M.S., 1987. Palaeomagnetic and geochemical studies on dolerite dykes from Tamil Nadu, India. Precambrian Res., 34: 291—310.

Dolerite dykes of the South Arcot, North Arcot and Salem districts of Tamil Nadu can be broadly divided into four major groups depending on their remanent magnetic directions. They are: (a) D = 86, I = —35, K = 400, α_{95} = 5, pole position at 189° E; 1° S; (b) D = 33, I = —48, K = 21, α_{95} = 18, pole position at 220° E, 38° N; (c) D = 133, I = —75, K = 134, α_{95} = 6, pole position at 54° E, 31° N; and (d) D = 294, I = —52, K = 18, α_{95} = 23, pole position at 49° W, 13° N.

VGP's of these dykes are in good agreement with the earlier Precambrian data available for Peninsular India, and their pole positions indicate that all these dyke events could be older than 1830 ± 200 Ma. Petrochemical studies suggest that the dykes are quartz-normative tholeiites and must have formed at shallow depths; their differentiation trends reveal that they were probably derived from two different magmatic sources. These dykes are transitional in character between an oceanic and a continental setting, which suggests that they intruded into thinned crust during an extensional event.

INTRODUCTION

Dolerite dykes of different ages and orientations are frequent in the South Indian shield, where they cut various Archaean crystalline rocks. Systematic studies of these dykes have revealed various igneous episodes in the shield. Very few dates are available on these dykes. The oldest dykes are dated at 2420 ± 246 Ma (Ikramuddin and Stueber, 1976, Rb—Sr whole rock) south of Bangalore, but others in western Karnataka have been dated at 2193 ± 45 Ma (K—Ar, Balasubrahmanyan, 1975). Sugavanam et al. (1978) have suggested four episodes of basic dyke emplacement on lithological grounds, ranging from 2100 to 1000 Ma. Afnassayev et al. (1964) recorded a K—Ar age of 1680 Ma for a dyke near Madras. Radhakrishna et al. (1986) reported K—Ar ages of 1668 ± 31 Ma and 1660 ± 25 Ma for a dolerite dyke in the Kerala region. Newer dolerites represent intrusive activity over an interval extending from c. 1600 to 900 Ma (Sarkar, 1980). Crawford and Compston

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(1973) report a Rb—Sr age of 980 ± 110 Ma for dolerites intruding the Cuddapah sediments.

Palaeomagnetic results on dyke rocks from a few places in the Indian Peninsular shield have been reported by different workers (Verma et al., 1968; Athavale and Verma, 1970; Hasnain and Quereshy, 1971; Anjanappa, 1972, 1975; Prasad and Reddy, 1972; Verma and Prasad, 1974; Kumar and Rao, 1979; Reddy and Prasad, 1979; Bhalla et al., 1980 a, b; Kumar and Bhalla, 1983; Venkatesh et al., 1984; Rao and Radhakrishna Murthy, 1985; Radhakrishna et al., 1986). Their studies have contributed to the evaluation of various aspects of igneous activity, the nature of the geomagnetic field, tectonic movements and geological correlations.

Palaeomagnetic data are not available so far on Tamil Nadu dyke rocks, and we have therefore initiated work on dolerite dykes from the South Arcot, North Arcot and Salem districts of Tamil Nadu. The aim of this study is to understand the various igneous episodes in this area and to correlate them with the other Precambrian formations and dyke rocks in the shield. Geochemical studies have also been carried out to supplement the palaeomagnetic results.

GEOLOGY AND SAMPLING

The dyke rocks have been collected from three different areas in Tamil Nadu and their detailed descriptions are given below. In total 90 oriented block samples from 19 sites representing 10 dykes were collected from these three areas using a Brunton compass and a precision spirit level. Sites with high magnetization intensity have been preferentially omitted as they affect the magnetic orientation. The samples were cored in the laboratory and cut into cylindrical specimens of standard size (2.5 cm diameter and 2.2 cm length), and a total of 543 speciments were subjected to palaeomagnetic studies.

(a) Kunnam, South Arcot

Two dolerite dykes (A and B) were sampled from this area (12° 06′ N, 79° 49′ E). These dykes are exposed for about 3 km as two prominent ridges from Semangalam Eri in the east to Perumbakkam Karadu in the west. They have a general trend of N 60° W and maintain a discordant relationship with the country rock (charnockite) which trends in a NE—SW direction. Two sites from dyke A and three sites from dyke B were sampled from different quarries. A minimum of three oriented block samples were collected at each site, and in total 26 samples were obtained. Figure 1a shows the sampling sites of the dykes.

(b) Tiruvannamalai, North Arcot

Five dolerite dykes (C, D, E, F and G) were sampled from different areas around Tiruvannamalai (Lat. 12°00′ — 12°11′ and Long. 78°53′ — 79°08′).

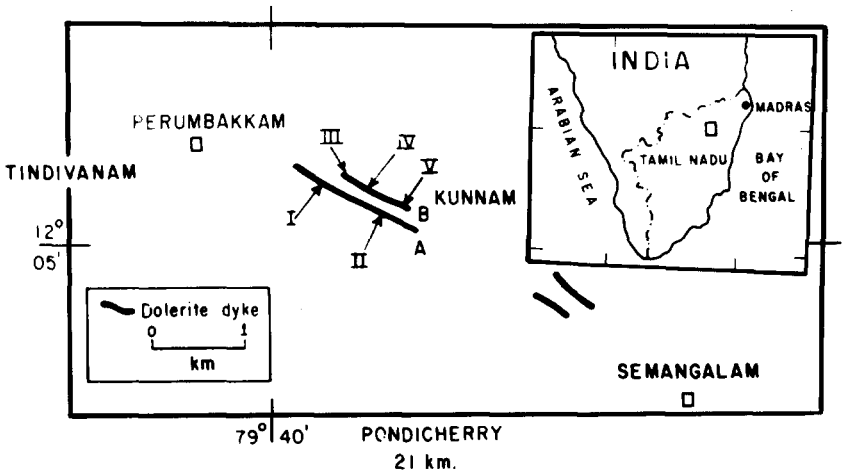
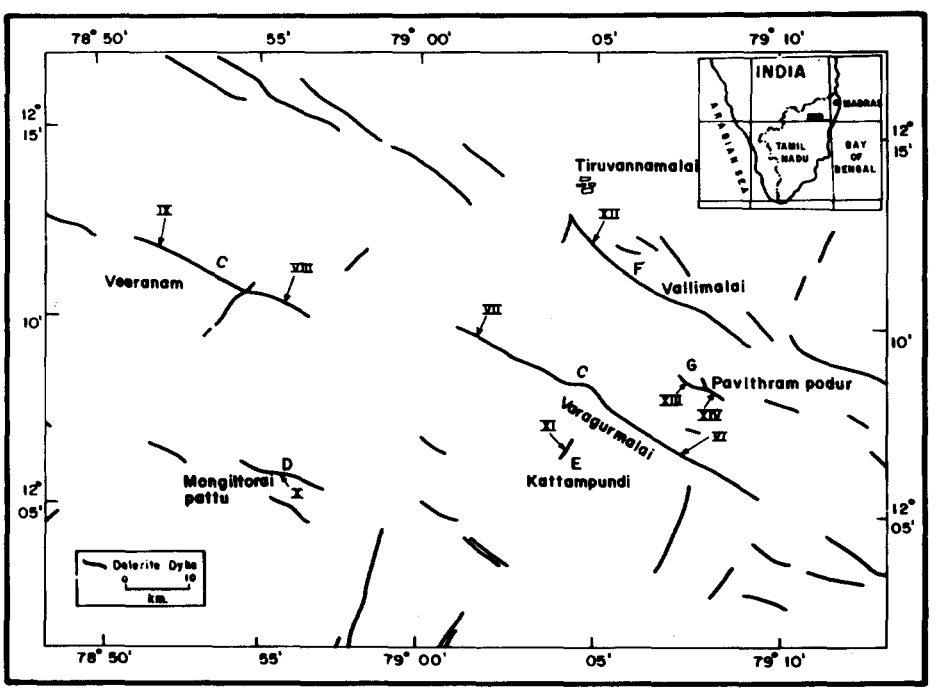
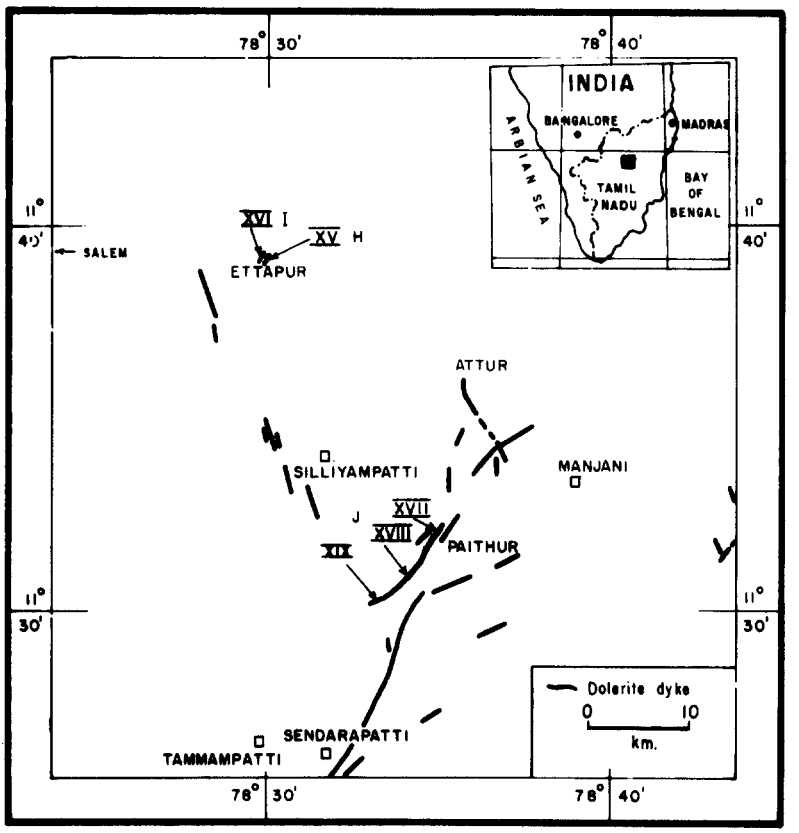


Fig. 1(a). Sketch map of the Kunnam area (Sotuh Arcot District, Tamil Nadu), showing palaeomagnetic sampling localities of dykes A and B. (b). Sketch map of the Tiruvannamalai area (North Arcot District, Tamil Nadu), showing palaeomagnetic sampling localities of dykes C, D, E, F and G. (After Sugavanam et al., 1978). (c). Sketch map of the Attur area (Salem District, Tamil Nadu), showing palaeomagnetic sampling localities of dykes, H, I and J.





These dykes have been mapped in detail by Sugavanam et al. (1978). Dykes in this area are of variable extension with vertical dips and they cut various hybrid gneisses and charnockites. Dykes C, D, F, and G trend N 50° W to N 70° W and dyke E trends N 20° E. Four sites were sampled from dyke C, whereas from the other dykes only one site at each dyke was sampled because of the dykes' limited extent and poor exposure. Figure 1b shows the sites of the dykes sampled. A minimum of three samples was collected at each site, which amounts to 40 samples from this area.

(c) Attur, Salem

Three dolerite dykes (H, I and J) were sampled from this area (Lat. 11°30′ — 11°40′ and Long. 78°28′ — 78°35′). The dykes generally trend N 30° E — N 55° E and cut the country rock (charnockite). A very long dyke (Fig. 1c) was sampled from Paithur quarries. A minimum of four oriented samples was collected from each site and in total 24 block samples were obtained from all of the five sites from these three dykes. Sampling locations are shown in Fig. 1c.

EXPERIMENTAL PROCEDURES AND RESULTS

The remanent magnetic directions and intensities of all the specimens were measured on an astatic magnetometer. Stability of remanence in these samples was tested by alternating field and thermal demagnetization techniques. For this purpose an A.F. demagnetizer with a two axis tumbling arrangement similar to that of Creer (1959) in design and a Schonstedt thermal demagnetizer, model TSD-1, were used. Susceptibility measurements (Likhite and Radhakrishnamurthy, 1965) and *Qn* ratios (Koenigsberger ratios) for all the specimens were also computed.

NRM measurements

NRM intensities vary from 0.26 to 85.95 A m^{-1}. The measured susceptibilities of the specimens vary between 4.65 and 108.11 ($k_{SI} = 4 \pi k_{emu}$). The computed *Qn* ratios (Koenigsberger) are between 0.16 and 72.59. The *Qn* ratio for a large part of the specimens is between 0.16 and 15.0. NRM directions of all sites show varying degrees of scatter. The NRM vectors for dykes A, B, H and I group well, while the remaining dykes showed considerable scatter in NRM directions. For dykes A and B (Kunnam) the NRM directions have ENE pointed declinations with shallow (both positive and negative) inclinations while dykes H and I (Attur) have NRM directions with ENE pointed declinations and shallow positive inclinations. NRM vectors with corresponding stable vectors (sample means) are shown in Fig. 2.

Cleaning techniques

All specimens were subjected to stability tests in progressively increasing alternating magnetic fields and temperatures to establish the primary remanent magnetic vectors in these rocks. A minimum of two specimens from each site was taken for alternating magnetic field and thermal demagnetization pilot studies and a total of 45 specimens were studied by A.F. demagnetization and 52 specimens by thermal demagnetization.

During A.F. demagnetization, the specimens were subjected to stepwise increasing peak fields of 2.5, 5, 10, 15, 20, 30, 40, 60, 80 and 100 mT. During thermal demagnetization, the specimens were heated at temperature intervals of 100, 200, 300, 400, 450, 500, 530, 560, 580, 600, 620 and 630 °C. Susceptibilities were measured before and after each heating step, and hardly any change in susceptibility values was observed, indicating that no chemical change occurred during the heating experiments.

The resulting vectors form a cluster in fields between 20 and 60 mT during A.F. demagnetization. The specimens, however, retain the same NRM vector during thermal demagnetization up to 600 °C, without much change in intensity and direction, after which the intensity drops and directions become erratic. Based on the pilot studies, batch fields were determined for each

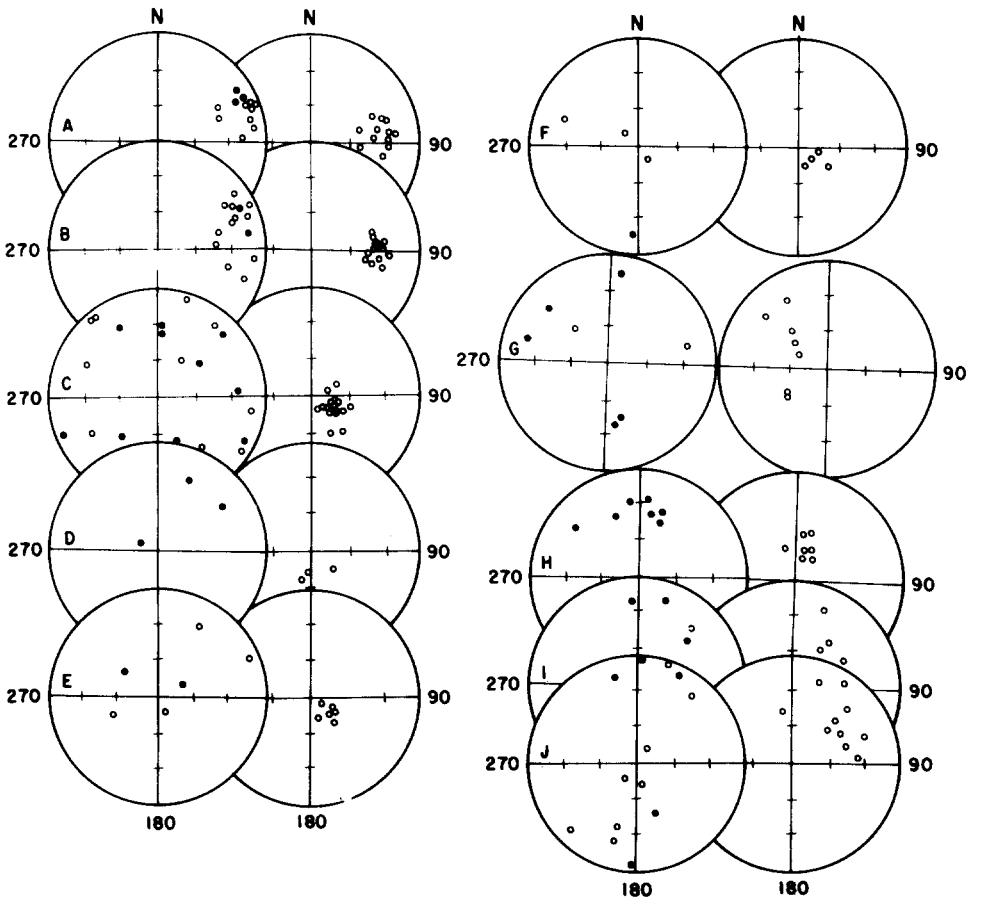


Fig. 2. NRM and stable vectors of samples from dykes A to J, plotted on stereographic projections, solid (open) circles are downward (upward) inclinations.

site and the rest of the specimens from that site were then demagnetized at the respective fields. A.F. demagnetization was found to reveal the stable vector in all the dykes. The specimens from dykes A and B showed similar remanent directions after both A.F. and thermal treatments. The response of typical pilot specimens to A.F. and thermal treatments are shown in stereographic projection in Figs. 3 and 4, respectively, and on orthogonal projection in Fig. 5. The dykes have been divided into four different groups (a, b, c and d) based on their magnetic directions. The response of the specimens to A.F. and thermal demagnetization of each individual group is given below.

Group (a) dykes, A and B (Kunnam)

The remanent vectors of this group have shallow negative inclinations with ENE declinations after both magnetic and thermal cleaning (Figs. 3 and 4).

Considering the better grouping of remanent vectors after A.F. cleaning than during thermal treatment, we consider the remanent vectors obtained from A.F. demagnetization to represent the primary magnetic directions for this group of dykes as given in Fig. 2.

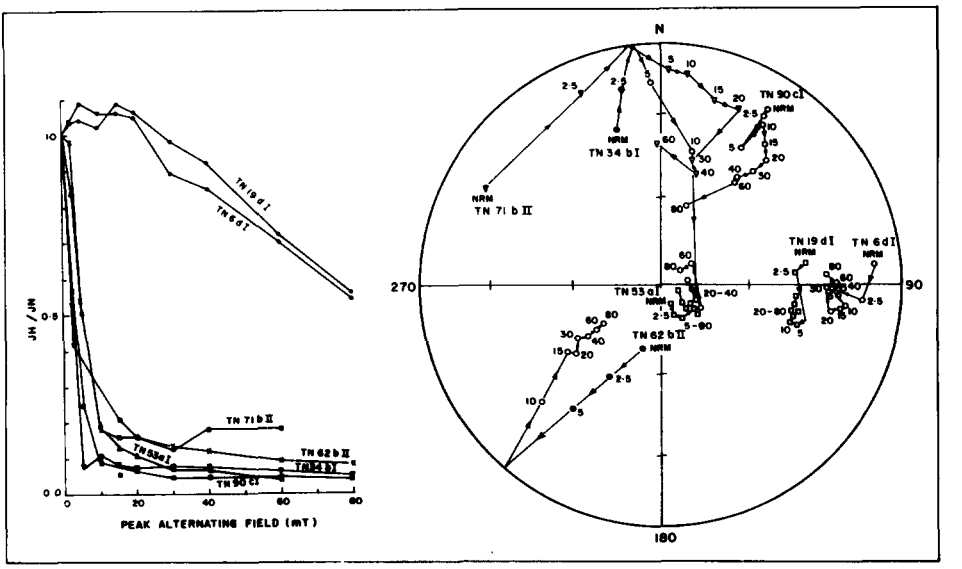


Fig. 3. Stepwise AF demagnetization data. Numbers next to symbols indicate corresponding peak alternating field in mT.

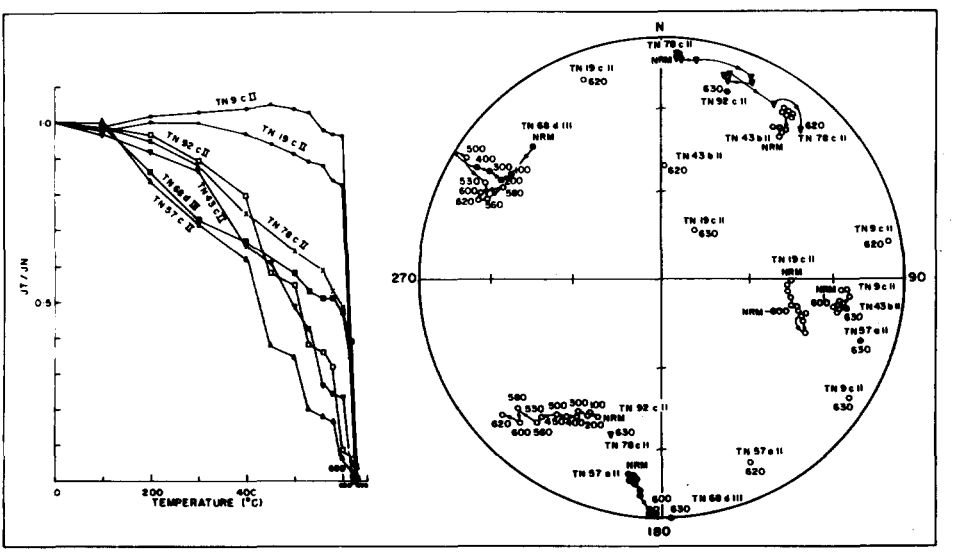


Fig. 4. Stepwise thermal demagnetization data. Numbers next to symbols indicate corresponding temperatures.

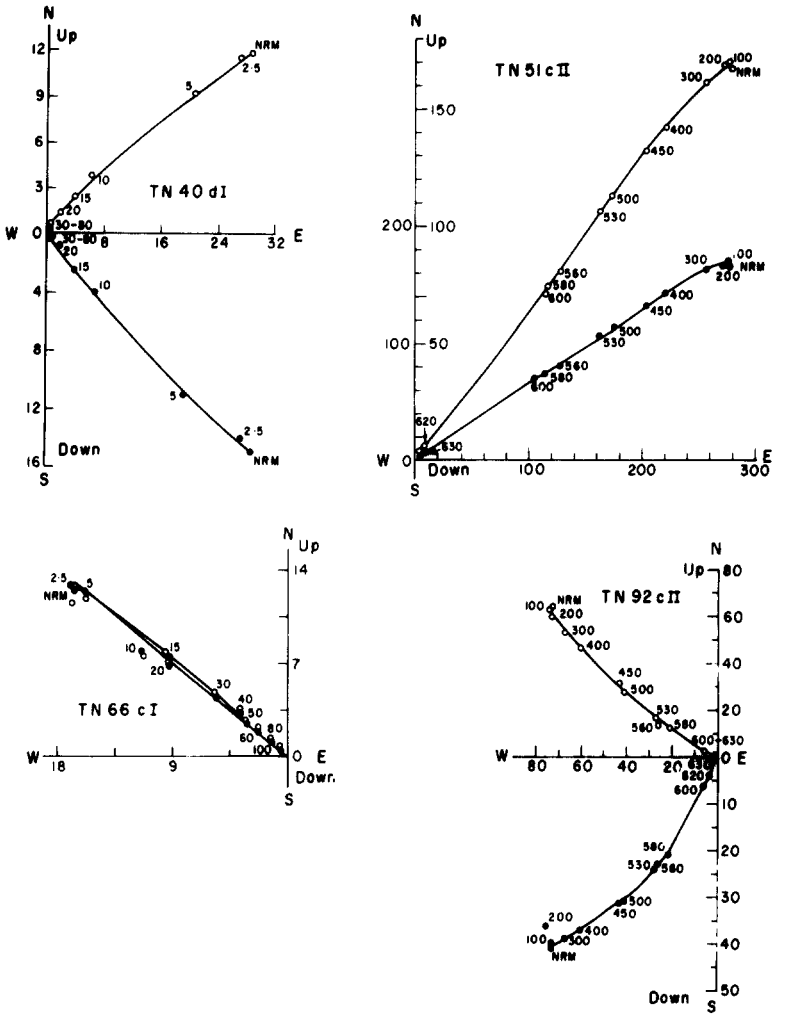


Fig. 5. Zijderveld diagrams for A.F. demagnetization (TN 40 bI, TN 66 cI) and thermal demagnetization (TN 51 cII, TN 92 cII). Solid symbols are end points projected on the horizontal plane, and open symbols are end points projected on the vertical plane. Numbers next to symbols indicate values of corresponding peak alternating field in mT and temperature in degrees Celsius used during demagnetization. Intensities are in units of A m^{-1}.

Group (b) dykes, H, I and J (Attur)

There is improvement in the remanent magnetic vectors from this group of dykes as a result of A.F. treatment. The NRM vectors, which are highly scattered, change to intermediate negative inclinations in the first quadrant (Fig. 2). Thermal treatment did not prove to be effective in this group of dykes (Figs. 4 and 5).

Group (c) dykes, C, D, E and F (Tiruvannamalai)

There are four dykes in this group. Except for dyke C, all of the other three dykes are represented by only one site, each with three to six samples. The NRM vectors from these dykes show large scatter which after cleaning changes to a very good grouping in the ESE quadrant with steep negative inclinations. Orthogonal projection (Zijderveld, 1967) for pilot specimen during A.F. demagnetization (Fig. 5) clearly shows a single stable primary vector after removal of weak secondary components. In the case of this dyke, A.F. treatment proved to be more effective in isolating the stable vector than the thermal technique. The remanent vectors obtained after A.F. treatment are shown in Fig. 2.

Group (d) dyke, G (Pavitram Podur)

This group contains one dyke with two sites showing dissimilar remanent magnetic directions with the other dykes. The majority of specimens have positive inclinations that later change to negative inclinations. Considering its typical and dissimilar remanent magnetic direction nature, this dyke has been named separately as Group (d).

Stable directions were obtained during A.F. batch cleaning and better grouping was observed in fields between 20 and 60 mT. Specimens were subjected to batch cleaning in these fields to obtain stable magnetic directions. Some specimens with high values of *Qn* ratios were found to be out of grouping with others after demagnetization and hence were rejected in the final computations. Fisher's (1953) statistics were used to evaluate the NRM means and stable primary means of each site and dyke, as well as groups of dykes. The data are listed in Table I along with other palaeomagnetic parame-

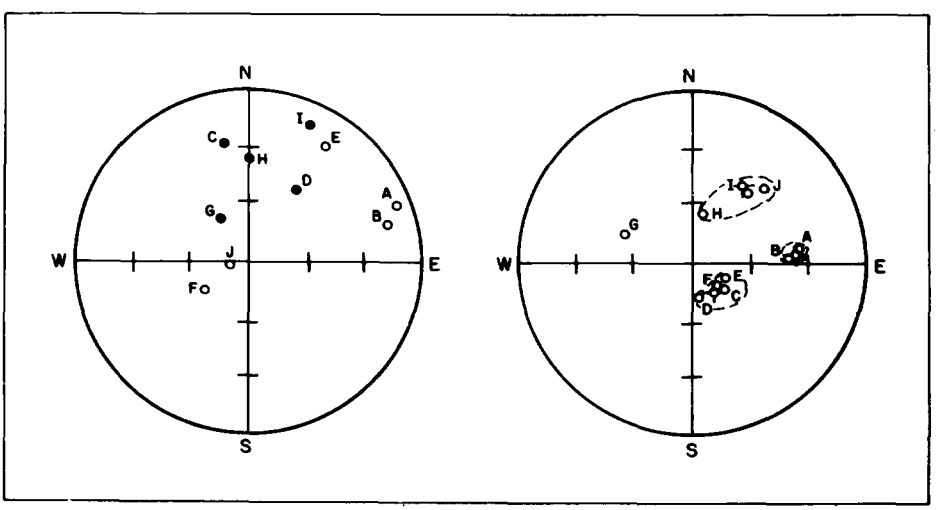


Fig. 6. NRM and stable treatment magnetic vectors of the dykes studied. Note the different clusters of remanent vectors after A.F. demagnetization.

TABLE I

Palaeomagnetic results of Tamil Nadu Dykes

Dyke	Site	N R M						Field (mT)	After AF Demagnetization					
		N	*n*	*D*	*I*	*K*	α_{95}		*N*	*n*	*D*	*I*	*K*	α_{95}
Group-a														
A	I	6	49	69	-5	17	14	20	6	25	78	-32	40	9
—do—	II	6	41	71	-10	17	14	40	6	17	89	-35	60	7
Dyke mean	—	—	—	—	—	—	—	—	—	—	83	-34	156	8
B	III	6	39	74	-25	26	11	20	6	20	87	-38	98	6
—do—	IV	5	35	87	-7	9	21	20	5	17	94	-33	100	6
—do—	V	3	17	63	-13	33	14	40	3	10	87	-37	230	5
Dyke mean	—	—	—	—	—	—	—	—	—	—	89	-36	345	4
Group-b														
H	XV	7	37	0	34	11	16	40	7	22	9	-63	52	7
I	XVI	4	22	24	11	6	29	40	4	15	36	-41	17	17
J	XVII	6	39	72	-61	1	53	40	5	16	55	-36	10	19
—do—	XVIII	4	22	17	-66	1	58	40	3	9	44	-36	31	14
—do—	XIX	3	17	233	6	1	75	40	3	10	32	-39	5	37
Dyke mean	—	—	—	—	—	—	—	—	—	—	44	-37	75	9
Group-c														
C	VI	6	34	318	-6	2	45	40	5	16	116	-75	127	6
—do—	VII	7	36	90	25	2	40	40	7	19	112	-77	87	6
—do—	VIII	3	16	240	9	1	92	60	2	7	151	-54	103	10
—do—	IX	4	18	23	10	2	51	20	3	9	102	-69	76	9
Dyke mean	—	—	—	—	—	—	—	—	—	—	126	-70	36	12
D	X	3	17	33	50	4	40	40	3	10	169	-74	32	14
E	XI	6	36	33	-19	1	54	30	6	16	115	-75	75	7
F	XII	4	25	236	-62	2	50	40	4	13	125	-75	58	9
Group-d														
G	XIII	4	23	106	44	1	63	40	4	11	275	-60	5	30
—do—	XIV	3	20	299	0	3	44	40	3	10	307	-42	15	21
Dyke mean	—	—	—	—	—	—	—	—	—	—	294	-52	18	23

POLE POSITIONS

Dykes	*N**	D_m	I_m	*K*	α_{95}	λ_p	L_p	λ_m	δ_p	δ_m
Group-a	2	86	-35	400	5	1 S	189 E	19 S	3	6
Group-b	3	33	-48	21	18	38 N	220 E	21 S	15	23
Group-c	4	133	-75	134	6	31 N	54 E	66 N	10	11
Group-d	1	294	-52	18	23	13 N	49 W	29 S	22	32

Groups a to d are the youngest to the oldest episodes. The approximate age relationships of these groups are discussed in the test. *N* = Number of samples, *n* = number of specimens, D_m = mean declination, I_m = mean inclination, *K* = precision parameter, α_{95} = radius of circle of confidence at 95% probability (Fisherian statistics were based on the specimen mean directions), *N** = number of dykes, λ_p and L_p = palaeocoordinates of pole, λ_m = ancient latitude of Nagpur (centrally located place m India), δ_p, δ_m = semi-axes of oval of 95% confidence.

ters. The dyke mean directions (Table I) provide the basis for classifying them into four dyke groups as shown in Fig. 6.

As a further check, a two stage treatment programme was adopted (not shown in figure) to isolate the primary component from any secondaries. The specimens were subjected to A.F. demagnetization in low fields followed by thermal demagnetization. However, this test did not yield any meaningful result as there is not as much change in the magnetic directions as there is in intensity.

PETROGRAPHY

The dykes are medium to fine grained with plagioclase feldspar (50 — 52%) and pyroxenes (28 — 37%) forming subophitic to ophitic relationships. In some samples chlorite is often observed as a reaction rim around pyroxene and forms at the expense of pyroxene. The augite shows marginal crystallization to hornblende and contains numerous small opaque inclusions. Clouding occurs in plagioclase feldspar to varying degrees and occurs to a lesser extent in pyroxene. The clouding takes the form of a pale-brown to black dust. Most of the plagioclase laths have an intensely clouded core that grades uniformly into a clear rim. Thickly clouded feldspars are seen in dykes A, B, H and I; the remaining dykes show feebly clouded to clear plagioclase. In the augite, the clouding concentrates near the grain margins particularly in areas adjacent to the large opaque grains. At high magnifications the cloudiness in many areas is resolved as numerous submicron inclusions. Several previous studies (Carstens, 1955; Armbrustmacher and Banks, 1974) suggest that the inclusions in the plagioclase are iron oxides, mainly magnetite. Hypersthene, biotite and quartz were also noticed in a few samples.

The opaques consist of magnetite and sulphides (pyrite/pyrrhotite) in order of abundance. Generally the opaque oxide grains are subhedral magnetite to coarse oxidation—exsolution intergrowths of magnetite and ilmenite grains with a few specks of sulphides. Subhedral to skeletal magnetite grains are found to be disseminated. Pyrrhotite is pinkish to yellow, pleochroic and anisotropic; pyrite is yellow, non-pleochroic and isotropic. Both minerals show medium to high reflectivity with cubic to subrounded grains.

GEOCHEMISTRY

Twenty samples from 10 dykes have been chemically analysed to see the relation between chemical composition and magnetization, their chemical variance with time and their magma character. Table II gives the results of the chemical analysis.

The overall chemistry of the dykes shows that they are quartz-normative tholeiites with SiO_2 varying between 49.0 and 54.5%. They are intermediate in tectonic setting between oceanic and continental tholeiites (Fig. 7). A plot of different oxides against the differentiation index (DI of Thornton and Tuttle, 1960) shows little scatter. There is a slight increase in SiO_2 with

TABLE II

Chemical analysis data of Tamil Nadu dykes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
SiO_2	51.50	52.10	52.20	54.50	52.90	53.40	51.30	50.40	51.30	49.90	50.40	52.60	51.90	53.20	54.10	49.00	49.90	49.70	49.70	51.40
TiO_2	1.18	0.98	1.09	0.89	0.90	1.27	1.20	1.23	1.29	1.37	1.42	1.43	0.82	0.75	0.70	1.61	1.57	1.04	1.18	1.09
Al_2O_3	14.70	15.10	14.60	13.90	13.40	14.40	13.90	12.20	12.40	12.20	13.30	13.60	14.30	13.80	12.70	15.20	14.50	14.80	14.70	14.80
Fe_2O_3	2.00	2.68	3.21	3.00	3.10	3.50	3.00	3.80	3.20	3.99	4.50	3.40	3.80	4.00	6.47	6.90	4.49	5.42	5.23	3.34
FeO	10.20	7.80	9.70	9.10	8.90	10.40	10.40	10.10	10.80	9.56	10.20	11.00	7.90	7.84	4.60	9.12	9.48	8.48	8.84	10.36
MnO	0.31	0.28	0.29	0.29	0.29	0.31	0.31	0.31	0.31	0.32	0.31	0.33	0.26	0.26	0.27	0.37	0.37	0.29	0.29	0.30
MgO	6.26	6.91	5.88	6.11	5.78	5.45	5.61	6.67	5.95	6.82	5.37	5.34	7.82	8.71	9.70	5.36	5.61	5.85	6.98	6.27
CaO	9.65	10.52	10.35	10.02	12.80	9.21	9.67	10.80	9.78	10.83	8.63	8.35	9.99	7.14	7.54	8.80	9.16	9.62	10.13	9.41
Na_2O	2.22	2.06	2.42	2.43	2.49	2.30	2.49	2.28	2.31	2.25	2.25	2.14	1.58	1.39	1.33	2.27	2.16	2.09	2.09	2.02
K_2O	0.43	0.39	0.41	0.36	0.38	0.47	0.50	0.39	0.43	0.39	0.51	0.54	0.35	0.58	0.50	0.33	0.36	0.37	0.46	0.35
P_2O_5	0.28	0.24	0.26	0.22	0.26	0.21	0.22	0.22	0.25	0.23	0.14	0.42	0.15	0.09	0.13	0.04	0.04	0.12	0.17	0.09
LOI	0.18	0.18	0.03	0.40	0.12	0.01	1.65	0.19	0.32	0.16	0.46	0.14	0.51	0.35	0.54	0.38	0.45	0.33	2.12	0.01
Total	98.91	99.24	100.44	101.22	101.32	100.93	100.25	98.59	98.34	98.02	97.49	99.29	99.38	98.11	98.58	99.38	98.09	98.11	101.89	99.44
(ppm)																				
Co	70	60	50	80	100	80	40	50	60	60	70	100	70	40	50	50	60	70	50	10
Cr	154	295	154	189	154	123	118	167	132	220	132	150	110	594	638	696	154	123	259	215
Ni	86	179	94	132	125	101	172	140	109	109	101	86	70	187	245	205	39	62	148	148
V	100	450	600	550	500	650	600	500	550	600	650	750	600	800	250	250	550	600	600	500
Cu	141	114	132	118	114	150	141	136	155	109	136	186	182	100	82	82	100	73	155	136
Pb	33	22	11	11	11	22	11	33	22	22	11	33	11	33	11	11	11	11	11	33
Zn	93	61	87	75	84	89	90	68	92	76	87	95	120	59	62	72	118	109	87	77
Sr	136	124	127	133	151	124	133	176	127	206	121	124	133	100	88	85	212	215	121	118

Index: dyke A = 1,2,3; dyke B = 4,5; dyke C = 6,7,8,9; dyke D = 10; dyke E = 11,12; dyke F = 13; dyke G = 14,15; dyke H = 16; dyke I = 17; dyke J = 18,19,20.

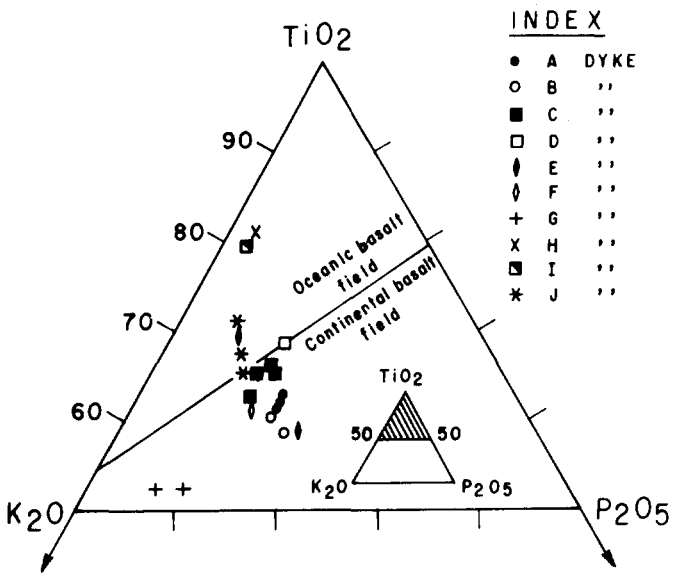


Fig. 7. Plot of dykes on TiO_2—K_2O—P_2O_5 diagram showing the position of oceanic and continental basalt fields (diagram after Pearce et al., 1975).

differentiation, which is accompanied by an increase in Na_2O and a decrease in MgO, CaO and Cr (Fig. 8). The data form clusters without any trend for some elements, the low DI values probably indicate little differentiation and a primitive magma source.

Except for samples from dyke G all other dyke samples show more or less the same differentiation trend, which suggests that they are variants of a single magmatic source. The behaviour of dyke G with respect to the above indices shows that it was derived from a separate magmatic source. Samples from this dyke G are less differentiated while samples from the other dykes show more differentiation, which suggests that dyke G could be more primitive in nature than the other dykes.

DISCUSSION

Fisherian statistics were computed at site level for all the accepted specimen directions and were subsequently followed by dyke means where two or more sites existed on each dyke, as can be seen in Table I. Based on the similarity in the remanent magnetic directions, the dykes were pooled into different groups assuming that they were of more or less contemporaneous events. This is the reason why the virtual geomagnetic poles, which corresponded to the stable directions from various dykes, were pooled as separate groups (Table I).

Stable magnetic directions were obtained for all 10 dykes. They can be

classified into four groups by means of their similarity in magnetic directions and virtual geomagnetic pole positions. Their mean directions and VGP's are listed in Table I. The VGP's for the different groups are plotted in Fig. 9 on a Molleweide's projection. This was drawn from the available Precambrian data that are listed in Table III, which were mainly obtained from dolerite dykes and some other Precambrian formations in the Indian Peninsular shield. Anjanappa (1975) also identified four episodes of igneous intrusions around Tirupati on the basis of field, petrological and palaeomagnetic studies. Based on the geological observations in Northern Tamil Nadu, Sugavanam et al. (1978) also suggested four major episodes of basic dyke emplacement. Among them one is a noritic dyke, two are metadolerites, which are around 2100 Ma in age, and one is an unmetamorphosed dyke of 1000 Ma age (these ages are based on lithological and structural data only). Palaeomagnetic data from Kumar and Bhalla (1983) also show that there are at least three different igneous episodes in Peninsular India.

Clouding in feldspar and pyroxene, the formation of reaction rims around pyroxenes and the appearance of chlorite and biotite is commonly seen in

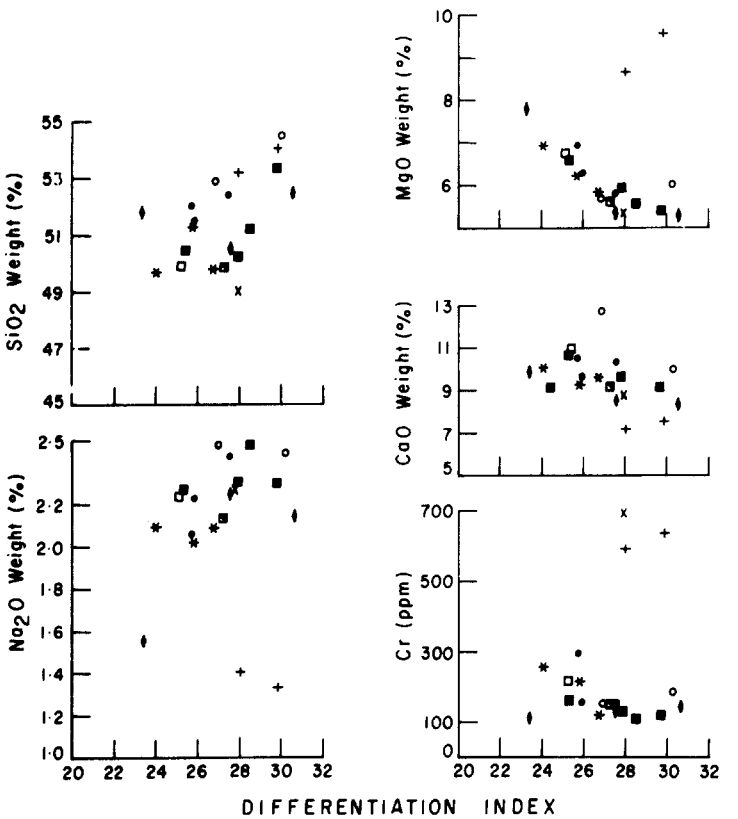


Fig. 8. Dyke compositions vs. Differentiation Index. Symbols as given in Fig. 7.

TABLE III

Palaeomagnetic data of dolerite dykes and some Precambrian formations in the Indian Peninsular shield

Sl. No.	Rock type	D_m	I_m	λ_p	L_p	λ_m	Reference
1.	South Arcot dykes (A and B)	86	-35	1 S	189 E	19 S	Present study
2.	Hunsur dyke	97	-14	8 S	172 E	6 S	Bhalla et al. (1980a)
3.	Karimnagar dyke	67	12	23 N	169 E	10 N	Rao and Radhakrishna Murthy (1980)
4.	Gwalior Traps	79	26	16 N	161 E	13 N	McElhinny et al. (1978)
5.	Gwalior Traps	78	35	19 N	155 E	19 N	Klootwijk (1974)
6.	Gwalior Traps	70	3	19 N	176 E	1 N	Athavale et al. (1963)
7.	Anantapur dykes	57	-8	31 N	182 E	4 S	Kumar and Bhalla (1983)
8.	Salem dykes (H, I and J)	33	-48	38 N	220 E	21 S	Present study
9.	Coimbatore dykes	38	-34	42 N	204 E	10 S	NGRI unpublished data
10.	Hyderabad dyke	44	-3	43 N	173 E	11 N	Verma et al. (1968)
11.	Kolar dykes	37	-18	47 N	197 E	2 S	Reddy and Prasad (1979)
12.	Hyderabad dyke	32	-16	49 N	205 E	5 S	Kumar and Rao (1979)
13.	Karimnagar dyke	40	-63	17 S	50 E	56 N	Rao and Radhakrishna Murthy (1980)
14.	Anantapur dykes	71	-72	3 N	47 E	61 N	Kumar and Bhalla (1983)
15.	Bangarpet dyke	129	-80	24 N	63 E	75 N	Bhalla et al. (1980b)
16.	North Arcot dykes (C, D, E and F)	133	-75	31 N	54 E	66 N	Present study
17.	Mysore dykes	141	-75	34 N	57 E	67 S	Hasnain and Quereshy (1971)
18.	Gwalior Traps	3	-74	4 S	77 E	66 N	Klootwijk (1974)
19.	North Arcot dyke (G)	294	-52	13 N	49 W	29 S	Present study
20.	Anantapur dykes	320	-34	38 N	68 W	19 N	Kumar and Bhalla (1983)
21.	Mysore dyke	329	-48	39 N	51 W	23 N	Hasnain and Quereshy (1971)
22.	Magnetite Ore (Unaltered)	310	25	42 N	5 W	—	Bhaskara Rao and Lakshmipathi Raju (1979)
23.	Kolar dykes	318	-6	45 N	30 W	2 N	Reddy and Prasad (1979)
24.	Kolar dykes	324	9	50 N	36 W	5 S	Reddy and Prasad (1979)
25.	Singhbhum dyke	330	-9	50 N	53 W	5 N	Verma and Prasad (1974)
26.	Tirupati dyke	349	-17	62 N	75 W	9 N	Anjanappa (1972)

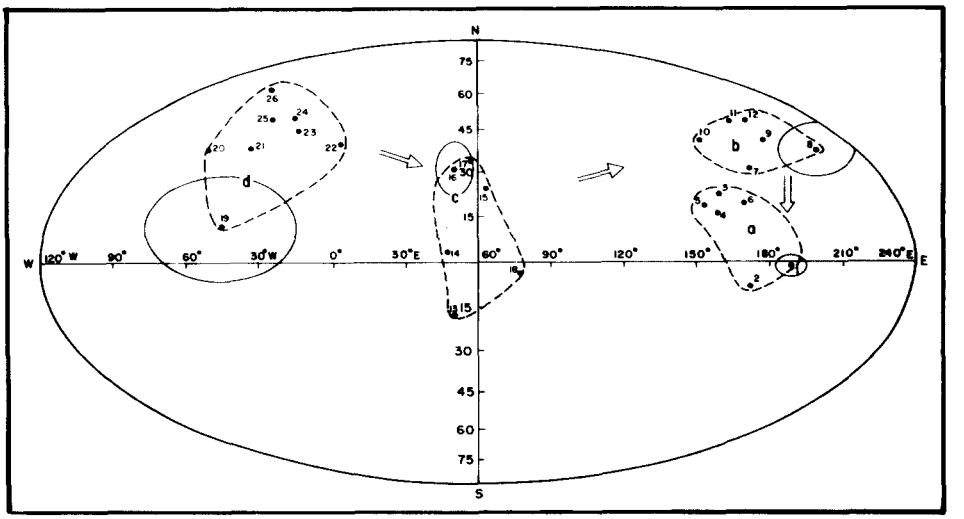


Fig. 9. Molleweide's projection showing the virtual geomagnetic pole positions of the data listed in Table III. The oval around pole positions of present study represents 95% oval of confidence.

the older dykes of the Indian shield. Several workers reported these features in dykes from various areas. Pichamuthu (1951, 1959) observed clouding of plagioclase feldspars in many dykes from the Dharwars, Nilgiris and several other places and attributed them to regional thermal metamorphism. In a separate note Pichamuthu (1967) noted two periods of metamorphism, that were responsible for the formation of two different generations of charnockite separated by a period during which numerous mafic dykes intruded. Sugavanam et al. (1978), Murthy (1980) and Bhaskara Rao (1953) also observed clouding in Tamil Nadu dyke rocks. Bhaskara Rao (1953) attributed clouding to deuteric effects at a later stage.

These dykes should be older than the sediments in the Cuddapah basin whose upper age limit is not more than 1700 Ma on the basis of lithological considerations (Murthy, 1979). There are a few radiometric ages on the dykes in the Indian Peninsular shield. Several workers (Foote, 1880; Hatch, 1902; Rama Rao, 1940; Janardhan, 1964; Anjanappa, 1975; Kumar and Bhalla, 1983) established Pre-Cuddapah ages for the dyke rocks from different areas in the Peninsula. Rb—Sr dating of lava samples from the lowest group of Cuddapah system shows that the age of the base of the system may be as great as 1700 Ma (Crawford and Compston, 1973). Pichamuthu (1959) assigned Pre-Cuddapah ages for the dolerite dykes with clouded plagioclase. Afnassayev et al. (1964) recorded a K—Ar age of 1680 Ma for a dolerite dyke intruding the charnockites near Madras. Sugavanam et al. (1978) recognised dolerite dykes from three different periods on the basis of lithological considerations and assigned ages of 2100—1600 Ma for the dykes with clouded plagioclase.

Palaeomagnetism of these rocks provides palaeo-positions of the regions at the time of their evolution and is an important tool to correlate the timing of tectonic events in the absence of reliable radiometric age data. In the present study an overall palaeomagnetic synthesis has been made for the available Precambrian data along with the present study data that is listed in Table III and is shown on a Molleweide projection in Fig. 9. From Table III and Fig. 9, it is evident that the magnetic directions of dykes from different parts of the shield fall into four groups, which probably represent different episodes of dyke intrusions in the Peninsula.

The remanent magnetic directions and pole positions of the Group (a) dykes are similar to those of the upper Gwalior Traps, which were studied by Athavale et al. (1963), Klootwijk (1974) and McElhinny et al. (1978) and dated at 1830 ± 200 Ma (Rb—Sr method, Crawford and Compston, 1970). These traps may date this group. Remanent magnetic directions and pole positions of the Group (c) dykes coincide with the lower Gwalior Traps (1830 ± 200 Ma) which Klootwijk (1974) reported. The similarity of the remanent magnetic directions of the Gwalior Traps (1830 ± 200 Ma) with these dyke groups suggest their age to be more or less contemporaneous with them. Considering the magnetic directions and pole positions, we have tabulated another group, i.e., Group (b) and placed it in the intermediate position between Groups (a) and (c). None of the Group (d) dykes is dated but Anjanappa (1972) speculated that the Tirupati dykes, with similar directions of magnetization to those of the Group (d) dykes, could have intruded at the closing stages of the Archaean. The chemical characters of dyke G suggest a primitive nature and thus it probably represents an earlier phase of igneous activity in the shield. These dykes might have been intermittently intruded into the high grade terrain of southern India and thus it appears that there are four episodes of dyke intrusions in this area. In this study, we could not attempt the polar wander path for these dyke groups as there is very little radiometric data for the palaeomagnetically studied rock groups in the Indian shield.

Chemically all these dykes are quartz-normative tholeiites. The oceanic to continental tholeiitic nature of these dykes with their low K_2O contents probably indicate that they were intruded when the crust was thin and during extensional tectonic events.

The low DI ratios (25 — 31) with high Ni and Cr contents suggest that the original magma was primitive and tholeiitic (Dietrich et al., 1978). Dyke G shows chemical characters that are distinctly different from the other dykes. It is also less differentiated than the other dykes indicating that it was generated from a separate magmatic source. All the other dykes show a more or less single differentiation trend with little variation in different oxides, which indicates their derivation from a common parent magma.

CONCLUSIONS

There are at least four phases of pre-Cuddapah dyke intrusion in the Tamil Nadu area, which forms an integral part of Peninsular India. Group (d)

probably represents the oldest dykes, while Group (a) dykes are the youngest. Groups (b) and (c) could be intermediate in age between (a) and (d). Secondary alteration in some of these dykes has not affected their stability of magnetization.

From the chemical data it is likely that these dykes formed from two distinct separate magma sources and the chemical differences between the dykes may be explained by their different depths of tapping and different *P—T* conditions during their intrusion. Their primitive chemical nature, along with their remanent magnetic data, suggest that these dyke groups probably represent earlier phases of igneous activity in the shield. Dyke G, which forms the Group (d), appears to have formed much earlier than the other dykes; all these dykes have intermittently intruded into the southern Indian Peninsular shield. These findings are corroborated by the magnetic data.

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